

Senior AI Systems Architect (Innovation Lab)

Role Overview

We are seeking a **Senior AI Systems Architect** to lead the design of autonomous, multi-agent frameworks. This is not a standard "implementation" role; we need a researcher-engineer who can move beyond single-LLM calls to build stateful, graph-based intelligent systems. You will be responsible for creating novel workflows that integrate disparate AI tools (LangGraph, Vector DBs, and Custom Tooling) into cohesive, production-ready architectures.

Key Responsibilities

- **System Design:** Architect and deploy multi-agent systems using **LangGraph** or equivalent orchestration frameworks.
- **Research & Innovation:** Conduct deep-dive research into emerging AI trends (e.g., Agentic RAG, Symbolic-Neural hybrids) and justify architectural decisions with evidence-backed reasoning.
- **Engineering Excellence:** Build robust, stateful backends using **FastAPI** or **Python**, ensuring modularity and scalability across domains.
- **Feature Discovery:** Identify and prototype "Original" AI features that solve complex organizational "Black Box" problems.

Technical Requirements

- **Core Stack:** Expert-level proficiency in **Python**, **LangChain**, and **LangGraph**.
- **Retrieval & Search:** Experience implementing advanced retrieval patterns (RAG) using **Vector Databases** and web-search APIs (e.g., **Tavily**).
- **Graph Orchestration:** Deep understanding of **nodes**, **edges**, **conditional routing**, and **state management**.
- **Evaluation:** Ability to design "Red-Teaming" agents to audit system outputs for bias, hallucinations, and technical accuracy.

Preferred Qualifications

- A strong public footprint on **GitHub** demonstrating contributions to multi-agent frameworks or open-source AI tools.
- Previous experience in "Systems Thinking" roles where you designed end-to-end pipelines rather than just modular features.